Introduction to Cryptography

Lecture 38: Applications and Requirements of Public-Key Cryptography

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Learning Outcomes

- Identify three broad public-key applications.
- Explain six public-key system requirements.
- Distinguish one-way and trapdoor one-way functions.
- Recognize common attacks and mitigations.

Three Major Applications

- Encryption/decryption: protect confidentiality.
- Digital signatures: authenticate signer and detect modification.
- Key exchange or establishment: agree upon shared keying material.

Confidentiality Scenario

Bob publishes PU_B. Alice sends

$$C = E(PU_B, M).$$

Bob recovers

$$M = D(PR_B, C).$$

The public key must be authenticated.

Signature Scenario

$$h = H(M), \quad \sigma = \mathrm{Sign}(PR_A, h)$$

Bob checks

$$\mathrm{Verify}(PU_A, M, \sigma).$$

A signature authenticates; it does not inherently encrypt the message.

Key Establishment

- Public-key methods can establish a symmetric session key.
- Key establishment may involve a key agreement protocol or a key encapsulation mechanism.
- The resulting key protects bulk data with efficient symmetric cryptography.

Six Requirements: Part I

1. Easy to generate a matched public/private key pair.
2. Easy to encrypt using the public key.
3. Easy to decrypt using the private key.

Six Requirements: Part II

4. Infeasible to derive the private key from the public key.
5. Infeasible to recover plaintext from public key and ciphertext alone.
6. For some systems, the keys support related operations in either direction.

The sixth property is optional, not universal.

One-Way Functions

$$y = f(x) \quad \text{easy to compute}$$

$$x = f^{-1}(y) \quad \text{computationally hard}$$

Cryptographic hardness is based on computational assumptions, not secrecy of the algorithm.

Trapdoor One-Way Functions

Trapdoor

Inversion is computationally hard without special secret information, but feasible with the trapdoor.

RSA uses a mathematical trapdoor related to the factorization of $n = pq$.

Threats to RSA

- Factorization and mathematical attacks.
- Weak randomness or badly chosen parameters.
- Timing and other side-channel attacks.
- Chosen-ciphertext attacks against raw RSA.
- Public-key substitution when certificates are not validated.

Why RSA Needs Padding

- Textbook RSA is deterministic.
- Raw RSA has algebraic properties that enable attacks.
- RSA-OAEP is a standardized encryption padding scheme.
- RSA-PSS is a standardized signature scheme.
- Never treat arbitrary “private-key encryption as a safe digital signature.

RSA Performance

- Public-key exponentiation is slower than bulk symmetric encryption.
- Repeated squaring reduces modular exponentiation cost.
- The Chinese Remainder Theorem can accelerate RSA private-key operations.
- A common public exponent is 65537.

Discussion Questions

1. Why must public keys be authenticated?
2. Which public-key application does Diffie–Hellman primarily support?
3. What makes a function a trapdoor one-way function?
4. Why is OAEP needed for RSA encryption?

Summary

Public-key cryptography supports

confidentiality • signatures • key establishment

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